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CLI

This guide provides examples of how to use the ZeroTier client-side CLI. The client CLI is currently only responsible for managing a specific node and its network membership. To control and configure networks you'll need to use the Central UI or the Central API.

PERMISSIONS

On Unix-like systems you may need to use `sudo zerotier-cli`, while on Windows you will need to use an `administrator` command prompt. If you see `missing authentication token and authtoken.secret not found (or readable)` you're likely trying to run `zerotier-cli` from an account with insufficient privileges.

TIP

For each command you can use the `-j` flag to get formatted JSON as an output.

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Node Status

Displays the status of the ZeroTier service, including version, node ID, and online status.

```
zerotier-cli status
```

```
zerotier-cli status
```

Node Status	Explanation
ONLINE	The client is able to communicate with the global root infrastructure.
OFFLINE	Unable to contact the root infrastructure but will continue retrying. This could take a few seconds. If it takes much longer, see Router Configuration Tips.
TUNNELED	Unable to communicate over UDP/9993, ZeroTier is now using a TCP fallback relay.

Example output

```
200 info abcdef1234 X.YY.ZZ ONLINE
```

List All Peers

Lists all peers in a more human-readable format. Alternatively you can use `listpeers` for a parser-friendly output.

```
zerotier-cli peers
```

```
zerotier-cli peers
```

Example output

zerotier-cli peers

```
$ zerotier-cli peers

200 peers
<ztaddr>   <ver>  <role> <lat> <link>   <lastTX> <lastRX> <path>
1b1c1dcf3 1.14.0 LEAF     15 DIRECT   799      4533     5.79.166.66/3
514cb0def5 1.14.0 LEAF     0 DIRECT   2731     2731     192.168.7721/
b6579fc8f6 1.14.2 LEAF    86 DIRECT   9598     9521     35.208.213.19
77ccdf1fa8 -      PLANET  79 DIRECT   24648    69739    185.190.103.6
```

LEAF is another device like a user's laptop, a server, or a network controller; a peer with the role PLANET is a root server that's part of our infrastructure. Seeing unknown peers is not typically a security concern because they are usually part of our infrastructure. If seeing an unknown peer with the role LEAF (ztaddr: b6079f73c6 , and you're joined to a network beginning with b6079f73c6, it may be one of zerotier's hosted network controllers

Join a Network

Joins the specified ZeroTier network using its 16-digit hexadecimal network ID.

```
zerotier-cli join <network ID>
```

```
$ sudo zerotier-cli join 8056c2e21c000001

200 join OK
```

Network Join Status Codes

Join Status	Explanation
OK	You're up and running on this network!
REQUESTING_CONFIGURATION	Negotiating with the network Controller. This could take about a minute. If it takes much longer, see Router Configuration Tips.
NOT_FOUND	Typo in the Network ID?
ACCESS_DENIED	You node needs to be Authorized on this network (via my.zerotier.com).
PORT_ERROR	See Port Error

Leave a Network

Leaves the specified ZeroTier network.

```
zerotier-cli leave <network ID>
```

```
zerotier-cli leave 8056c2e21c000001
```

List All Networks

Lists all networks that the node is currently a member of.

```
zerotier-cli listnetworks
```

```
zerotier-cli listnetworks
```

Example output

```
200 listnetworks
<nwid> <name> <mac> <status> <type> <dev> <ZT assigned ips>
200 listnetworks 8056c2e21c000001 COMPO-NET ee:7b:f9:ab:a4:a7 OK PRIVATE
200 listnetworks 8056c2e21c000002 LANPARTY 86:59:c6:6b:e0:81 OK PRIVATE
```

Get a Network Setting

Retrieves the value of a specific network setting for the given network ID. Settings used with `get` and `set` commands may include property names from the JSON output of `zerotier-cli -j listnetworks`

```
zerotier-cli get <network ID> <setting>
```

```
zerotier-cli get 8056c2e21c000001 mtu
```

Example output

```
2800
```

Set a client-side setting

Sets a specific client-side setting that does not affect the rest of the network or nodes. Settings that can be used here are `allowManaged`, `allowGlobal`, `allowDefault` and `allowDNS`. To learn more about these settings see here.

```
zerotier-cli set <network ID> <setting>
```

```
zerotier-cli set 8056c2e21c000001 allowGlobal true
```

List Moons (Federated Root Sets)

Lists all federated root sets (moons) the node is aware of.

```
zerotier-cli listmoons
```

```
zerotier-cli listmoons
```

Join a Moon via Any Member Root

Joins a federated root set (moon) using its world ID and seed.

```
zerotier-cli orbit <world ID> <seed>
```

```
zerotier-cli orbit <world ID> <seed>
```

Leave a Moon

Leaves the specified federated root set (moon).

```
zerotier-cli deorbit <world ID>
```

```
zerotier-cli deorbit <world ID>
```

Generate a dump file

Generate a debug dump file with device and network information for troubleshooting. This file can be useful when talking with ZeroTier technical support staff.

```
zerotier-cli dump
```

```
zerotier-cli dump
```

CAUTION

The `dump` command creates a file on your desktop with information about your device and networks. Some of this information (Physical IP addresses, public identities, etc) may be sensitive.

Notes on Settings

Settings used with `get` and `set` commands may include property names from the JSON output of `zerotier-cli -j listnetworks`. For example:

```
zerotier-cli get <network ID> ip6plane
```

```
zerotier-cli get <network ID> ip6plane
```

Example output

This command retrieves the 6PLANE address assigned to the node.

Display all available commands

Displays the full set of client commands.

```
zerotier-cli -h
```

```
$ zerotier-cli -h

ZeroTier One version 1.14.2 build 0 (platform 1 arch 2)
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Usage: zerotier-cli [-switches] <command/path> [args]

Available switches:
  -h                    - Display this help
  -v                    - Show version
  -j                    - Display full raw JSON output
  -D<path>              - ZeroTier home path for parameter auto-detect
  -p<port>              - HTTP port (default: auto)
  -T<token>             - Authentication token (default: auto)

Available commands:
  info                  - Display status info
  listpeers             - List all peers
  peers                 - List all peers (prettier)
  listnetworks          - List all networks
  join <network ID>     - Join a network
  leave <network ID>    - Leave a network
  set <network ID> <setting> - Set a network setting
  get <network ID> <setting> - Get a network setting
  listmoons             - List moons (federated root sets)
  orbit <world ID> <seed> - Join a moon via any member root
  deorbit <world ID>    - Leave a moon
  dump                  - Debug settings dump for support

Available settings:
  Settings to use with [get/set] may include property names from
  the JSON output of "zerotier-cli -j listnetworks". Additionally,
  (ip, ip4, ip6, ip6plane, and ip6prefix can be used). For instance:
  zerotier-cli get <network ID> ip6plane will return the 6PLANE address
  assigned to this node.
```

Tags: client | administration | tutorial | examples

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